LAB-12

(1)

step 1: first we will initialize a (0) value to a variable.

step 2: we will initialize a loop of 'if' condition where we'll see if the initialized value which we kept is as zero (0), is less or equal than 10.

step 3: we will echo or print the output which will see or runs a loop for number 0 ~~as~~ first.

step 4: After completing the ~~upper~~ above loop we will increment the value by +1 so next the value which gets into a loop will be 1 and till 10 it reaches the loop breaks because the condition was less or equal to 10.

(2)

step 1: first initialize a range of value & and store it in a variable i = (0-50) → Example.

step 2: Initialize a loop of 'for' condition where evaluate if the current value in loop is modulo by 2.

step 3: If the remainder is equal to the zero, the condition is true and the number will print the value.

step 4: If the remainder is not equal to the zero, the loop will skip that value and will go to the next value

step 5: The loop will be terminated once the value of 50 gets evaluated and the values will be printed.

(3)

Step 1: We will read the number which ~~is~~ of the user will input.
Step 2: Initialize two variable, 'num' in which the input number will get stored and another variable which starts from 1.
i = 1

Step 3: We will set up an 'until loop' with the condition to terminate when the 'i' is greater than 15.
(until loop terminate as the condition turns True)

Step 4: Inside the loop, calculate the product of value which the user will input and the current value of 'i'. (Product = $((num * i)))
Step 5: Print the expression for ~~p~~ multiplication table.
($num x $i = $product)

Step 6: Once the product is executed for value 1 then we will increment the value of i by +1. (i = ($i + 1))

Step 7: The loop will end once the value of i becomes 16.

(4)

Step 1: Create text file called nums.txt with some number in the same line.
Step 2: Read & Store those number into a variable array called 'numbers'.
Step 3: We will print all the numbers in the file to check once the value is doubled.

Step 4: Initialize a loop in which each values goes through and during that time the numbers will get calculated by multiplying the current number by 2.
(original number * 2)

Step 5: Print the new numbers which are doubled.
Step 6: The loop will terminate as all the array elements get doubled.

(5)

step 1: Define a function called "divide" that take ~~two~~ argument one which ~~the~~ will be dividend and another will be divisor.

step 2: Inside the function "divide" all the variable should be declared local. (local divident, local divisor, local remainder, local quotient)

step 3: If the divisor which the second argument is equal to the zero then, print an error message, which will not divide the value zero and will exit the function.

step 4: If the divisor is not equal to zero, we will calculate the quotient upto 2 decimal places using awk.

step 5: Then after we will calculate the remainder using modulous operator (%)

step 6: Print both the quotient and the remainder.

(6) step 1: Define a function called "Maximum". which accepts two numbers as an arguments.

step 2: Inside the function will put a condition where we will see or compare the two numbers, which is greater. if the num1 is greater than num2 then it will print num1 otherwise it will print num2 as maximum

step 3: Outside the function give two numbers to execute and see which one is greater.

(7)

Step 1: Define a function which accepts the directory path as an argument.

Step 2: Make a variable and it would be defined as a local variable.

Step 3: We will check if the directory exists or not and if does not then we will create a directory

Step 4: Outside the function we will execute a directory to check if it exists or not and if it does not it will create a directory.